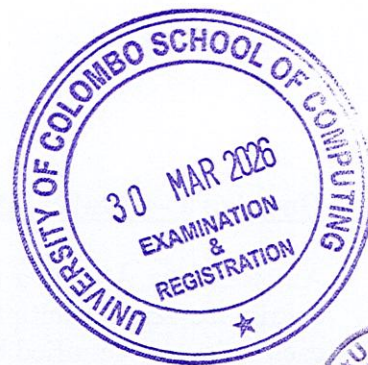


University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester II - UCSC AY22 [held in March/April 2026]

SCS1309 Database Management Systems

(Two (2) Hours)

Answer ALL questions

Number of Pages = 12

Number of Questions = 4

To be completed by the candidate

Index Number:

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Important Instructions to Candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number **CLEARLY** on each and every page of this Question paper.
- IV. This paper consists of **4** questions in **12** pages (including the Cover Page).
- V. **Answer ALL questions.**
- VI. Calculators and any electronic device capable of storing and retrieving text, including electronic dictionaries, smart watches and mobile phones **are not allowed**.
- VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

Question No	Marks
1	
2	
3	
4	
Total	

Index No:

Question 1 – Total 40 Marks

AutoRent Pvt. Ltd. is a nationwide car rental company that manages a large fleet of vehicles distributed across multiple branches. The company maintains detailed records of its customers, each uniquely identified by a CustomerID, along with their name, email address, multiple phone numbers, and residential address.

Customers may rent vehicles multiple times, and each vehicle may be rented by many different customers over time. Each rental transaction is identified by a RentalID. Whenever a vehicle is rented by a customer, the rental start date and rental end date are recorded. Each rental transaction is associated with exactly one customer and exactly one vehicle.

The company's fleet consists of different types of vehicles. Each vehicle is uniquely identified by a VehicleID and includes information such as registration number, model, manufacturer, year of manufacture, and daily rental rate. Some vehicles are categorized as luxury vehicles, which include features such as luxury level and complimentary service. Other vehicles are classified as commercial vehicles used for transporting goods, with features such as load capacity and special permit. Not all vehicles fall into these categories, and each vehicle may belong to at most one vehicle type.

AutoRent operates several branches located in various cities. Each branch is uniquely identified by a BranchID and has a branch name and contact number. Vehicles are assigned to branches; a branch may have many vehicles, but each vehicle is assigned to exactly one branch at any given time.

Each branch employs several staff members, including managers and service personnel. Every staff member is uniquely identified by a StaffID and has details such as name, position, and phone number. Each staff member works for exactly one branch, while a branch may employ multiple staff members.

For operational purposes, the company maintains records of vehicle maintenance activities. Each maintenance record includes the maintenance date, service type, and cost, and is associated with a specific vehicle. A vehicle may undergo multiple maintenance activities over time.

In addition, vehicles may have multiple insurance policies during their operational lifetime. Each insurance policy is uniquely identified by a PolicyNumber and contains details such as the insurance provider, coverage amount, and expiry date. Each insurance policy is associated with exactly one vehicle.

AutoRent intends to develop a database management system to systematically track customers, rental transactions, vehicle details and types, branch assignments, staff information, maintenance records, and vehicle insurance details. The system is intended to support efficient data management, operational control, and reporting across all branches nationwide.

(a) Draw an Enhanced Entity–Relationship (EER) diagram showing the necessary entities, attributes along with primary keys, relationships, and specializations for the above scenario. Specify the cardinality and participation constraints for each relationship set. Clearly state all assumptions, if any.

[20 Marks]

Index No:

- (b) Map the EER diagram developed in (a) above to a relational data model. Underline the primary key of each relation and explicitly indicate any foreign keys by marking them with (FK).

[20 Marks]

Question 2 – Total 10 Marks

Brew & Byte Cafe uses a digital loyalty system. They track the Customers, the Drink Menu, and the Orders placed. Consider the relational schema below:

Customer (CustomerID, Name, LoyaltyPoints)

Drink (DrinkID, DrinkName, Category, Price)

Order (OrderID, CustomerID, DrinkID, OrderDate, Quantity)

- (a) Provide the Relational Algebra Expression to list the names of all customers who have more than 500 loyalty points.

[1 Mark]

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- (b) Provide the Relational Algebra Expression to list the *Name* of the customer and the *OrderDate* for every order involving a 'Latte' as a drink name.

[2 Marks]

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- (c) Provide the Relational Algebra Expression to list drink names and order dates for drinks that cost more than 500 LKR.

[2 Marks]

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- (d) Provide the Relational Algebra Expression to calculate the total number of orders and total revenue per drink *Category*. Rename the final columns to *Drink_Category*, *Total_Orders*, and *Total_Revenue*.

[3 Marks]

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Index No:

- (e) Provide the Tuple Relational Calculus (TRC) expression for listing the customer names and their corresponding order dates for all orders placed.

[2 Marks]

Question 3 – Total 20 Marks

Assume you are the Database Architect for the sports platform at *Nexus-Global* company. Currently, the company uses a single flat file to track Tournament Matches, Pro-Players, and Sponsorship Deals.

Consider the following Unnormalized Form (UNF) relation.

Tournament_Master (*MatchID*, *PlayerID*, *MatchDate*, *PlayerName*, *TeamName*, *TeamLogo*, *GameTitle*, *PrizePool*, *SponsorName*, *SponsorCategory*)

The business rules related to the *Tournament_Master* relation are listed below.

- Rule 1: Each *MatchID* is associated with exactly one *GameTitle* and one *PrizePool*.
- Rule 2: A *PlayerID* corresponds to one *PlayerName*, one *TeamName*, and one *TeamLogo*.
- Rule 3: Each *TeamName* is associated with exactly one *SponsorName*.
- Rule 4: Each *SponsorName* belongs to one *SponsorCategory*.

- (a) Identify and list all Functional Dependencies (FDs) derived from the given business rules. Use the notation $X \rightarrow Y$.

[4 Marks]

FD1 based on Rule 1:

FD2 based on Rule 2:

Index No:

FD3 based on Rule 3:

FD4 based on Rule 4:

(b) Identify the primary key(s) of the *Tournament_Master* relation.

[1 Mark]

(c) Classify each of the functional dependencies (FD1–FD4) from Question 3 - (a) into Full Functional Dependencies, Partial Functional Dependencies, and Transitive Dependencies.

[3 Marks]

Full Functional Dependencies:

Partial Functional Dependencies:

Transitive Dependencies:

(d) Decompose the relation into **Second Normal Form (2NF)**. Clearly underline the Primary Keys for each new relation.

[6 Marks]

Index No:

- (e) Decompose the 2NF relations from Question 3 - (d) into **Third Normal Form (3NF)**.
Clearly underline the Primary Keys for each new relation.

[6 Marks]

Question 4 – Total 30 Marks

The Administrators at the Disaster Management Center use the *Ditwah Reporting Tool* to generate impact summaries for specific regions or dates. The tool retrieves data from three related tables that collectively capture geographical locations, weather observations, and disaster impact assessments.

In the *Location* table, the *loc_type* attribute classifies each location's terrain as *Inland* or *Coastal*. The *Observation* table records weather data, including *WindSpeed* (in kilometers per hour, KPH) and *Rainfall* (in millimeters, mm) during the observation period. The *Impact* table stores damage assessments (in millions) and human impact indicators, such as the number of displaced persons, deaths, homeless individuals, and injured persons, based on observations.

Index No:

Location

Loc_ID	loc_name	loc_type	Latitude	Longitude
801	Passara Hill	Inland	6.1395	80.1063
802	Batticaloa Coast-north	Coastal	7.7102	81.6924
803	Trincomalee Harbor	Coastal	8.5689	81.2330

Observation

Obs_ID	Loc_ID	Obs_Time	WindSpeed	Rainfall
501	801	2025-11-28 10:00	85.00	450.50
502	802	2025-11-28 11:30	70.00	310.00
503	803	2025-11-28 13:00	65.50	540.75
504	801	2025-11-28 14:00	95.00	560.00

Impact

ReportID	Obs_ID	PropertyDamage	Displaced	Dead	Homeless	Injured
9001	501	12.5	50	18	140	45
9002	502	83.2	27	3	76	22
9003	503	150	30	41	65	39
9004	504	29.5	15	12	42	18

Note: Answers written in any standard SQL dialect (e.g., Oracle, PostgreSQL, MySQL, MS SQL Server) are acceptable.

- (a) Write a SQL statement to retrieve the location name and location type of all records from the *Location* table and order the results alphabetically based on the location name.

[2 Marks]

- (b) Write a SQL statement to list the details of all observations where the recorded rainfall exceeds 400 mm.

[2 Marks]

Index No:

- (c) During a cyclone emergency, the Disaster Management Center frequently runs the query in Question 4-(b). Write a SQL statement to create an **index** named *idx_rainfall* on the *Rainfall* column of the *Observation* table to speed up this search.

[3 Marks]

- (d) Write a SQL statement to display each location type and the total number of observations recorded for that type. Rename the calculated column as *ObservationCount*.

[4 Marks]

- (e) Write a SQL statement to list locations where the total property damage exceeds *100 million*. Rename the calculated column as *TotalDamage*.

[4 Marks]

Index No:

- (f) Sensors sometimes malfunction and send negative values for rainfall. To maintain data integrity, write a SQL statement to create a **trigger** named *CheckRainfallValue*. This trigger should fire before a new record is inserted into the *Observation* table. If the *Rainfall* being inserted is less than 0, the trigger should automatically set that value to 0.00 before the record is saved.

[5 Marks]

- (g) Write a SQL statement to create a **view** named *ImpactSummary* that displays the following information for each location:

- Location details: loc_name, loc_type
- Observation time
- Impact information: PropertyDamage, Displaced, Dead, Homeless, and Injured persons.

[5 Marks]

Index No:

- (h) Write a SQL statement to create a **stored procedure** named *GetHighRainfall* that accepts a minimum rainfall amount as a parameter and returns the *Loc_ID*, *ObservationTime*, and *Rainfall* for all records that exceed that minimum amount.

[5 Marks]

*****THE END*****